

MATHWORKS HACKATHON 2026

Personal Fitness Tracker

Activity Recognition & Personalized Health Insights via Machine Learning

Team

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The Challenge

Build a smartphone-based fitness tracker from raw sensor data

From raw motion to actionable insights

Phone accelerometer is the only sensor — no smartwatch, no heart rate strap. **Can a phone in your pocket understand what you're doing — and how hard?**

15

recording sessions

3 users × 5 activities × 10 Hz sampling

01

Collect

labeled multi-sensor data across activities

02

Engineer

discriminating features from raw signals

03

Classify

with machine learning, validate honestly

04

Translate

into personalized health narratives

Three Users, Three Bodies

Different builds → different physiological outputs

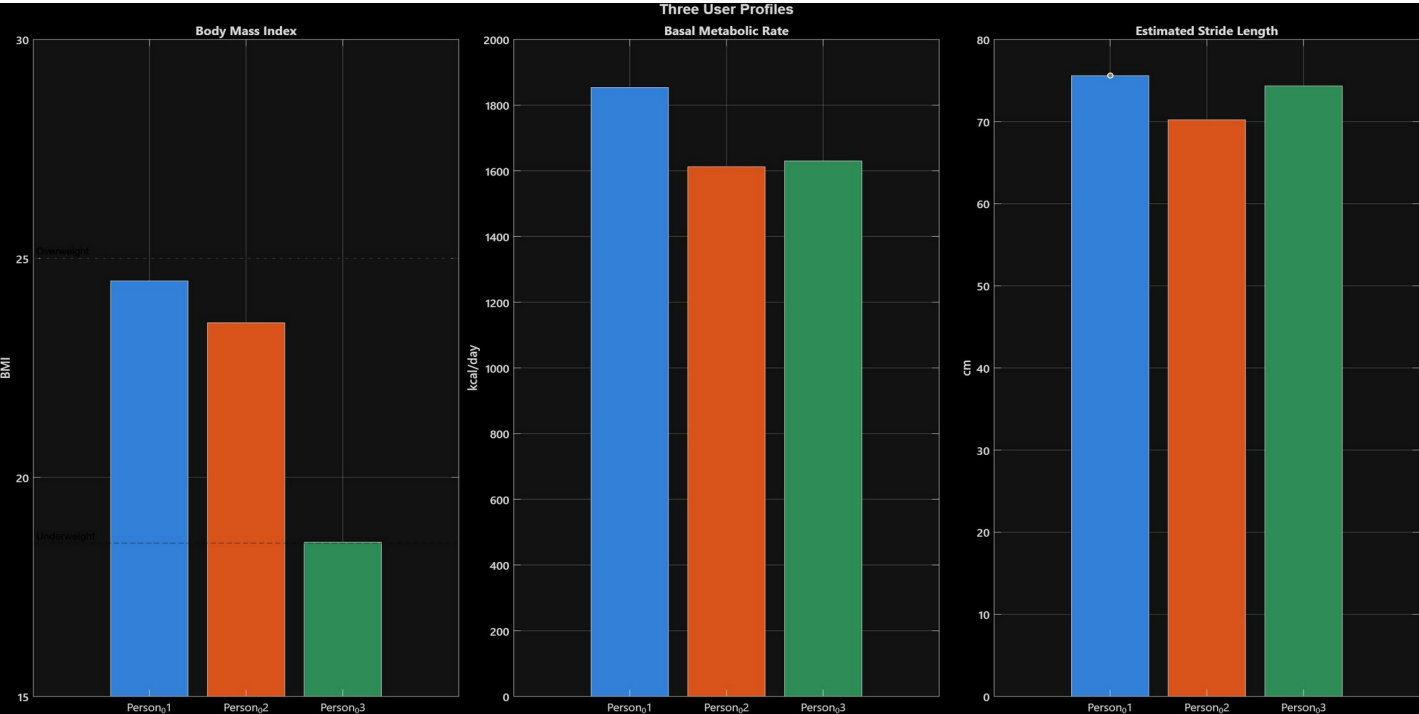


Figure: BMI, BMR, and stride length per user (anthropometric derivation)

Person_01
82 kg · age 23
BMI 24.5
stride 76 cm

Person_02
68 kg · age 27
BMI 23.5
stride 70 cm

Person_03
60 kg · age 20
BMI 18.5
stride 74 cm

Same exercise, different calorie burn — that's the story.

Data Collection: Activity Signatures Are Visible

Each activity has a distinct acceleration fingerprint

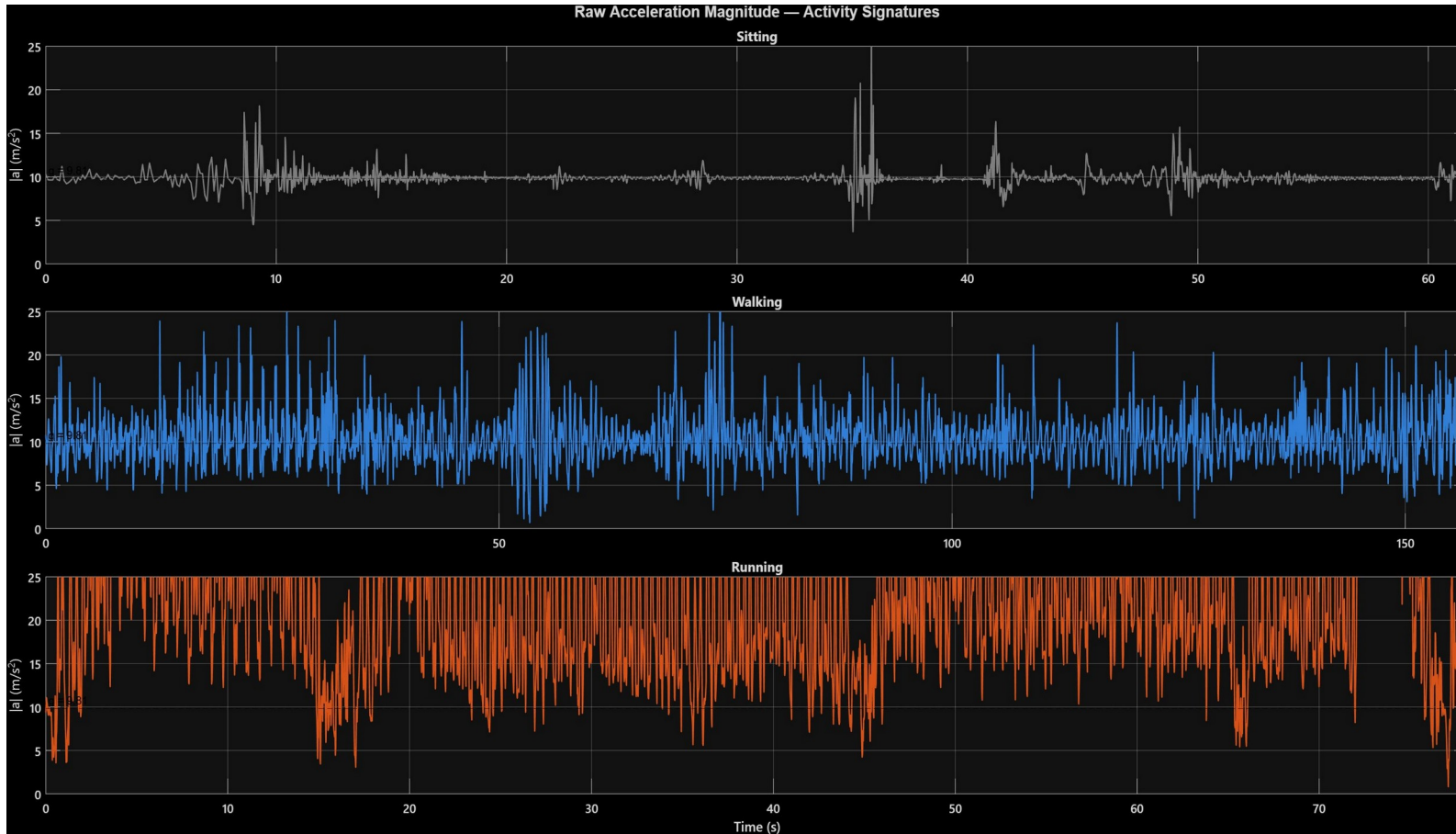


Figure: Raw acceleration magnitude — Person_01 across three activities

Setup

MATLAB Mobile Log mode

10 Hz sampling rate

Front pocket placement

4 sensors: accel · gyro · orientation · GPS

5 activities × 3 users

Visible insight

Sitting

Near $g = 9.81$, low variance

Walking

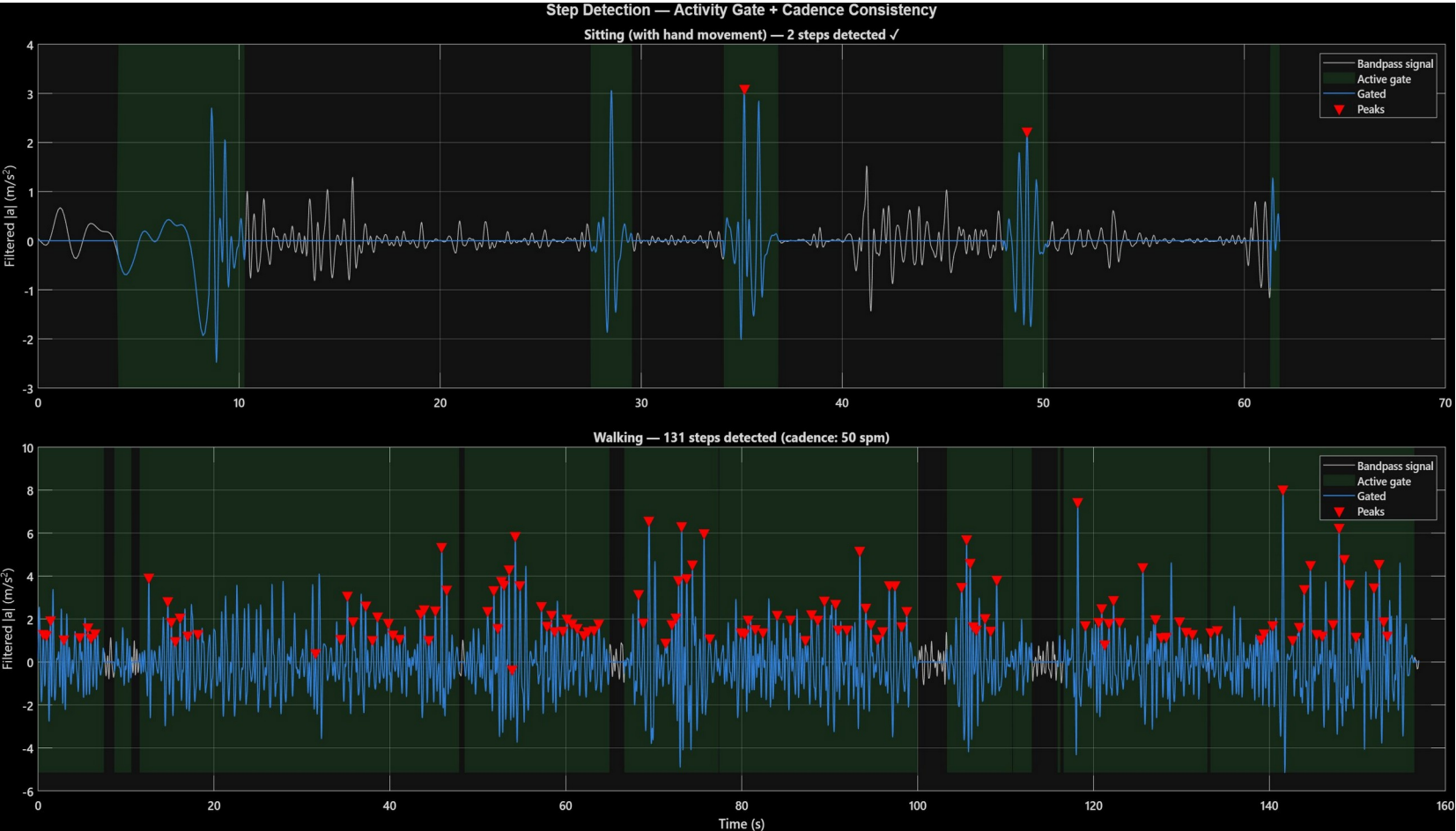
Regular ~ 2 Hz oscillation

Running

High amplitude, ~ 3 Hz peaks

Step Detection: Don't Be Fooled by Hand Shaking

Three-layer filter rejects non-rhythmic motion



3-Layer Filter

1

Bandpass

1–4 Hz cadence range only

2

Activity Gate

skip low-energy windows

3

Rhythm Check

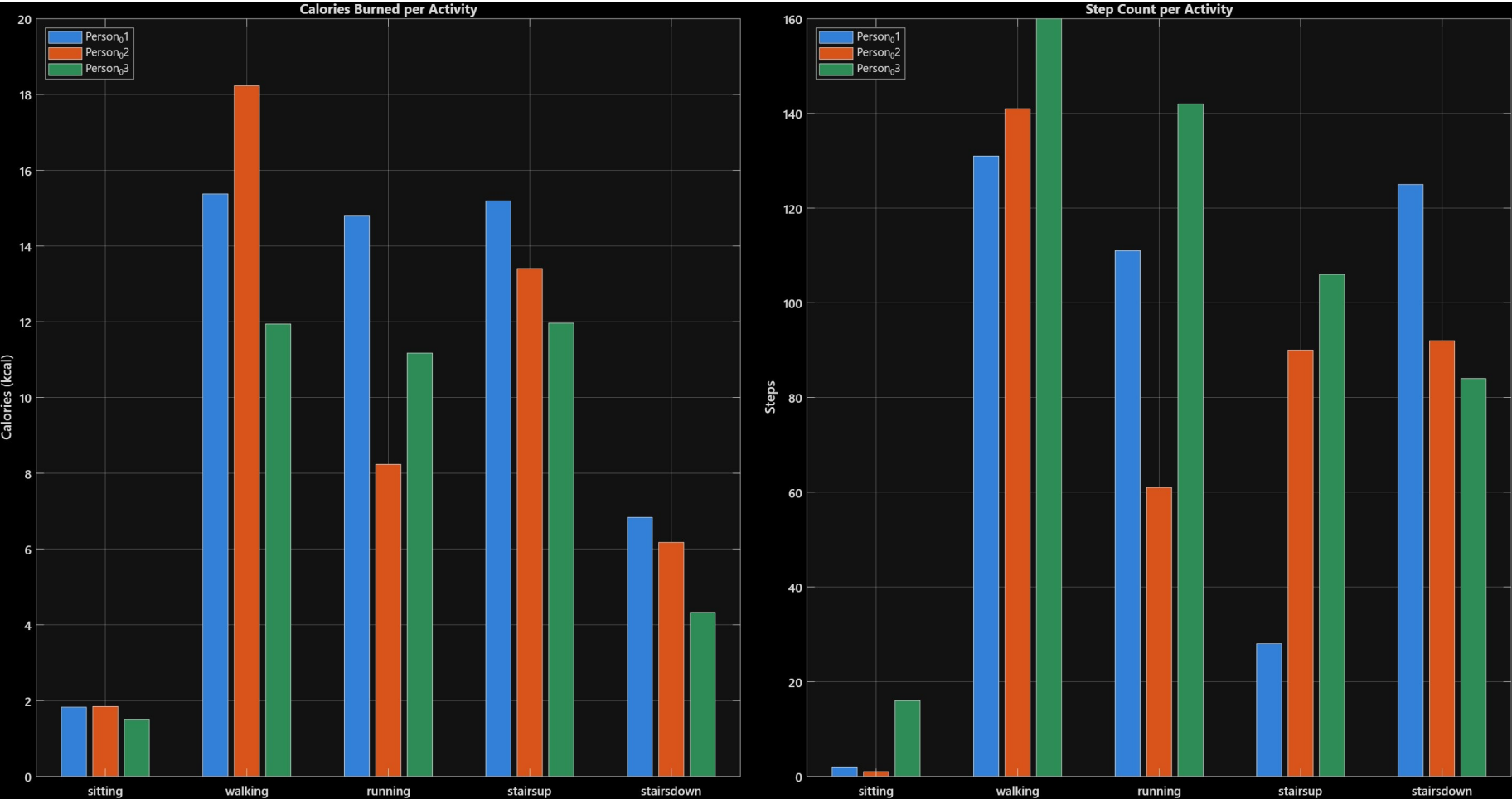
CV < 0.35 between peaks

Sitting → 2 steps ✓

Walking → 131 steps ✓

Per-Activity Metrics

Identical exercise, different physiological cost



+37%

more calories

Person_01 vs Person_03,
same walking session

MET-based formula

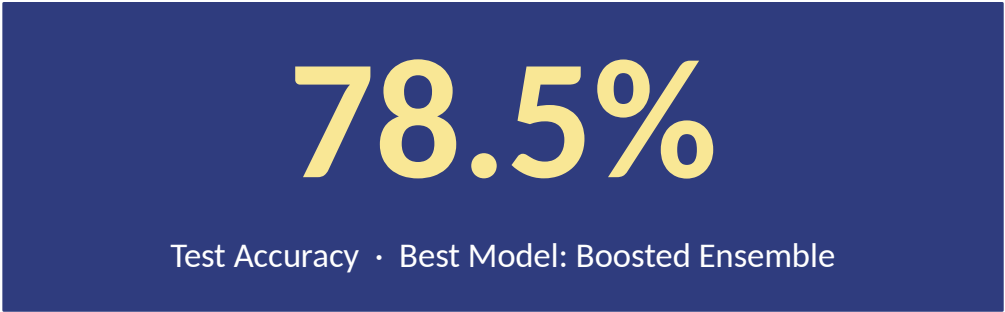
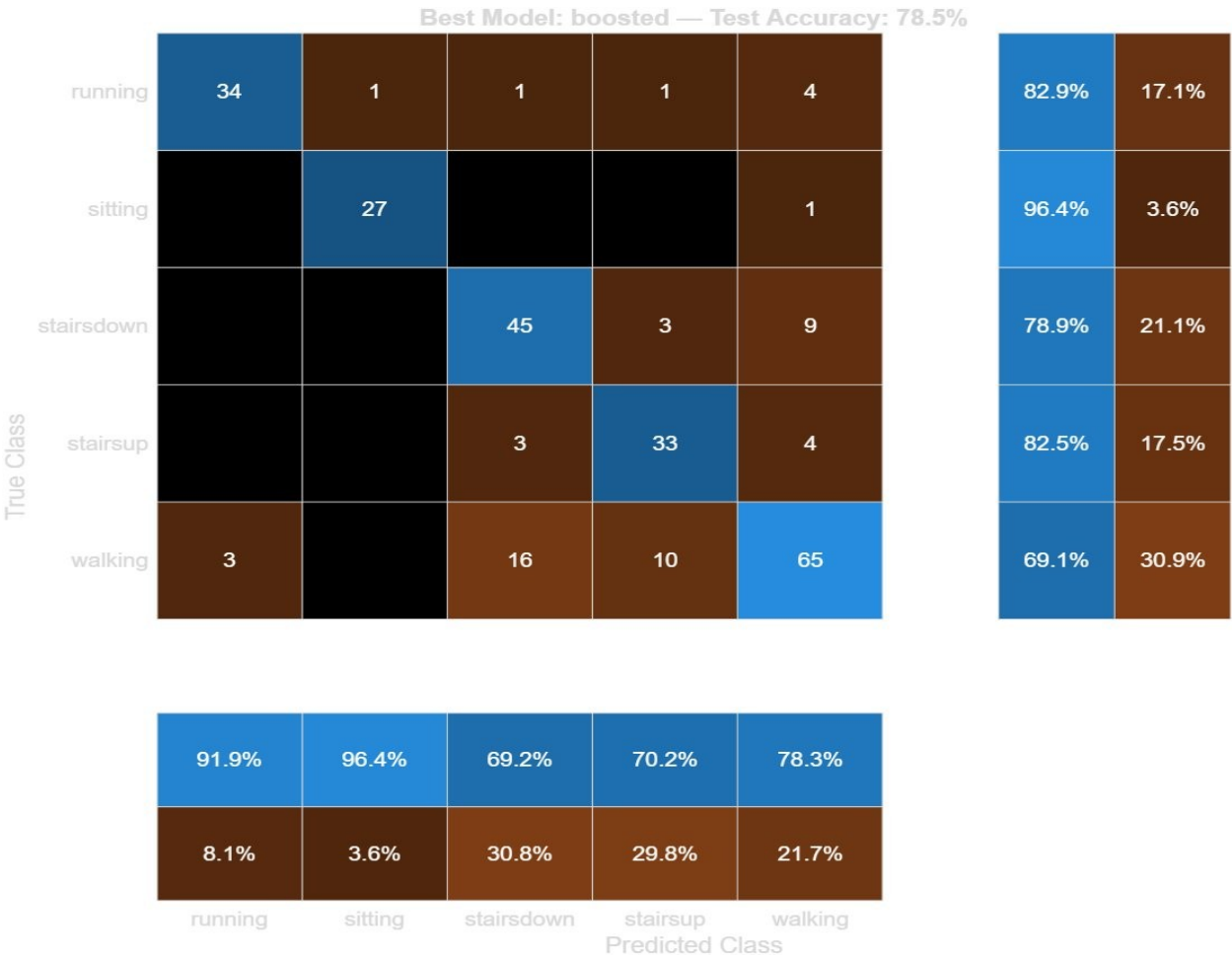
$$\text{kcal} = \text{MET} \times \text{kg} \times \text{hr}$$

ACSM 2011 Compendium

Body weight is the
calorie multiplier

Machine Learning: 78.5% Accuracy

Gradient-boosted trees on 23 engineered features



Pipeline

23 features from 2-second sliding windows

Per-axis std/RMS, magnitude statistics,
frequency bandpower, autocorrelation peak

4 models compared

KNN · Bagged · Boosted · Subspace KNN

Inverse-frequency class weights

Compensates walking-dominant data

Honest Test: Can It Generalize to a New User?

Train on 2 users, test on the 3rd — the hard test

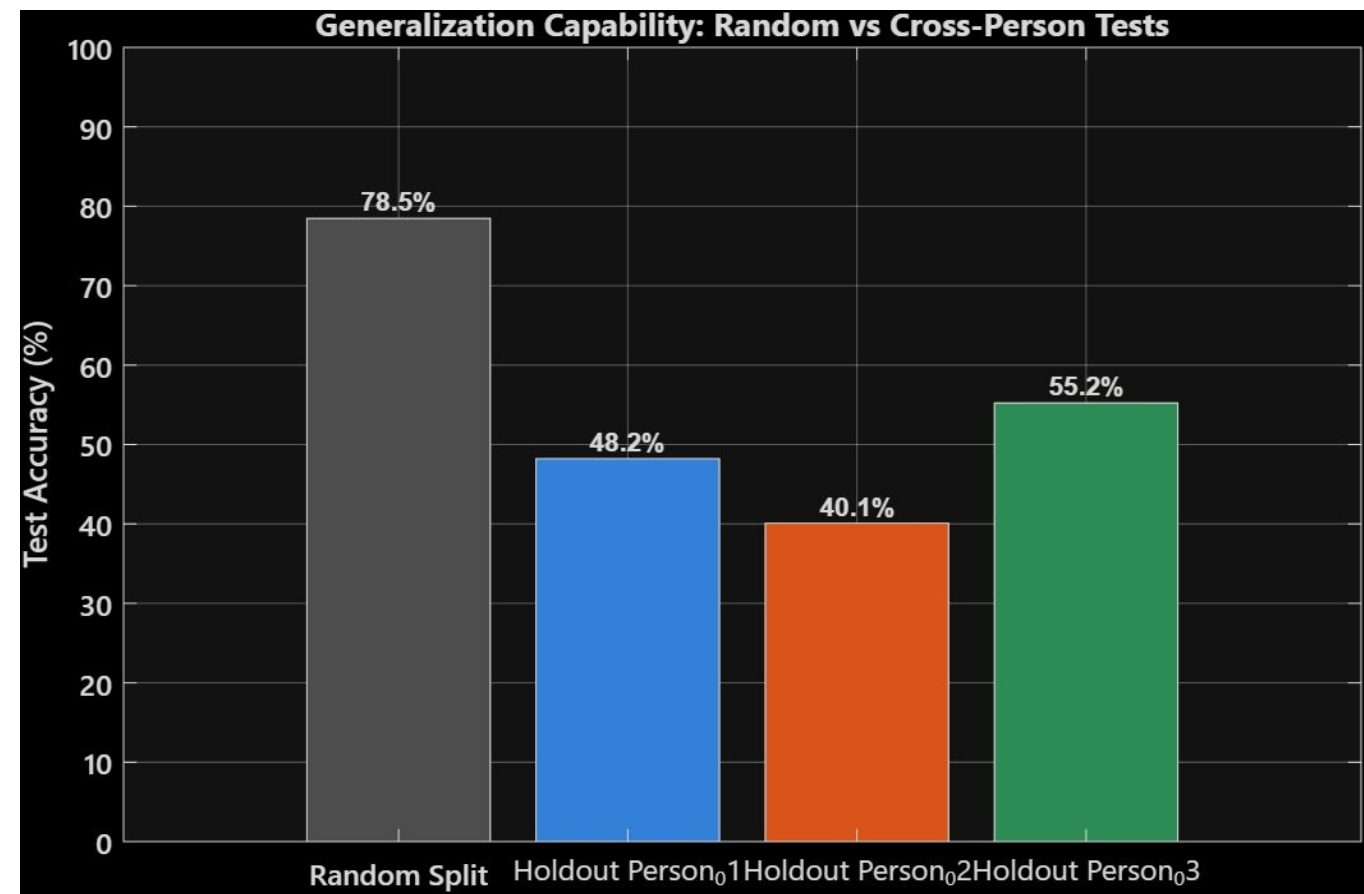


Figure: Random split (78.5%) vs cross-person holdout (40–55%)

What this tells us

The drop is real — and informative.

With only ~470 windows per person, the model learns both:

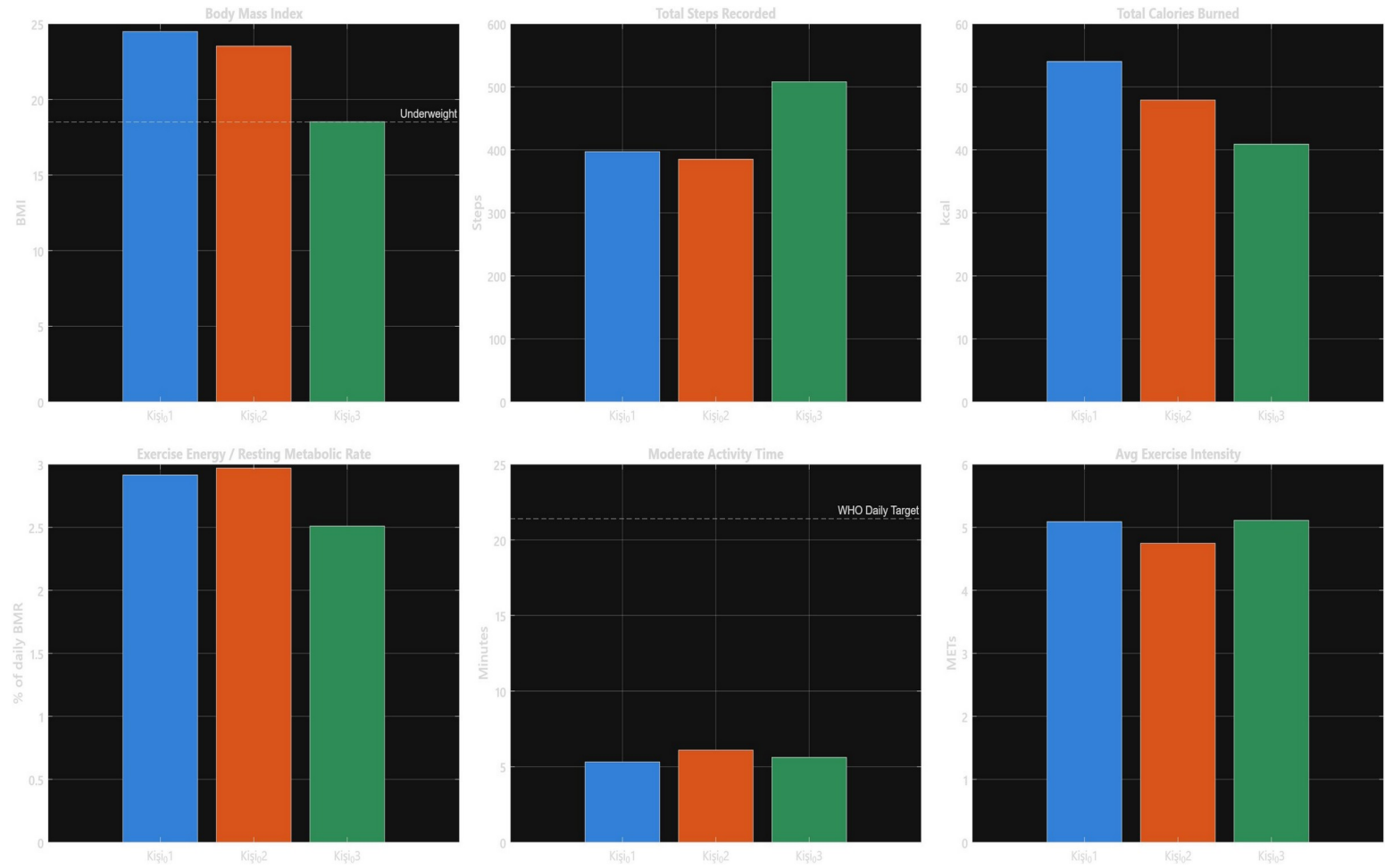
- ✓ Universal activity patterns
- ✗ Person-specific gait signatures

Production path

- Larger training sets per user
- Transfer learning across users
- On-device personalization

Personalized Health Insights

The tracker doesn't just count — it speaks to each user



Person_01

397 steps · 5.1 METs

Highest calorie output — body weight matters

54 kcal

Person_02

385 steps · 4.7 METs

Longest GPS distance (712 m walking)

47.9 kcal

Person_03

508 steps · 5.1 METs

Most steps — highest cadence efficiency

40.9 kcal

"Each user's energy expenditure, in their own terms."

Key Achievements

From raw sensor data to a personalized fitness narrative

15

labeled sessions

3 users × 5 activities

23

engineered features

time + frequency domain

78.5%

classification accuracy

boosted ensemble model

5

fitness metrics per user

steps · kcal · METs · WHO · BMR%

Three users · Five activities · One pipeline that knows the difference.

A fitness tracker that doesn't just count — it personalizes.

Thank you · Questions?